

Summary of Break-Through Energy Technologies

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1. Introduction and Background	
1.1. Background to this Briefing Document	

This summary document was prepared for the purpose of providing general background information about energy-producing technologies that are outside the usual main-stream consideration or awareness. This document is based on nearly 17 years of research by Dr. Greer, Dr. Loder and other members of the The Orion Project, who have been studying energy generation and propulsion technologies. We have identified a number of new energy technologies that within 6 months to several years, given proper funding, may be ready for marketing. A number of these technologies were presented at a briefing for the US Senate EPW in 2001 at the request of the Chair of the EPW, Senator Bob Smith (R-NH) (Loder, 2001).

Because many of these technologies are at the design or prototype stage, they will need funding for the necessary research to bring them out to the public. **To do this, it will be necessary to bring these candidate technologies to a stage of real commercial potential by developing what has been termed the “robust demonstrator”. This demonstrator would be a reproducible, self-powered, reliable device producing useable amounts of power and causing no pollution.**

1.2. Technology Categories and Background

We have identified a number of advanced technologies to replace the need for oil, gas, coal, and nuclear power, thus eliminating pollution and the geo-political instability resulting from the existing systems.

These technologies fall into three broad categories (examples are listed in Section 4.0):

- 1. Transitional technologies.** These are devices or processes that enable existing energy sources to be much more efficient, reduce pollution or clean up existing environmental hazards. An example would be add-on devices to retrofit cars and trucks with a technology that significantly increases fuel economy.
- 2. Electromagnetic energy devices that are over-unity.** These include generators, solid-state devices and magnetic plasma devices that perturb the zero point energy field (or quantum vacuum flux field) to tap electromagnetic energy and create usable power, without a net external fuel source or pollution.
- 3. Advanced propulsion systems that use electromagnetic fields to alter the mass and space around an object to allow for lift and drive.** These “electrogravitic” systems will enable travel to be pollution free, rapid and without the use of fossil fuels of any type or use of surface roads.

Heretofore, most efforts focusing on these technologies have been classified projects that have refused to allow widespread application and use due to various corrupt special financial and energy interests. Engineers, scientists, and inventors have either been absorbed into these so-

called ‘black’ projects or worked alone, without adequate R and D funding, support, strategic security or a professionalized process.

1.3. Development of Supportive Policies and Strategies

In addition to the science and engineering aspects of developing these technologies, there will need to be active development of commensurate policies and strategies in the following areas:

- **Application of these technologies in the developing world and the granting of such technologies to abate widespread poverty and environmental decay in those regions;**
- **Understanding the financial, economic and industrial implications of these technologies and determining how to best transition from a carbon-based industrial society to a clean, so called ‘free energy’ society, with minimal disruption;**
- **Determination of the national security and international security implications of these technologies and how to best empower existing governmental and international institutions to ensure a smooth transition to a new economy and geo-political order;**
- **Development of higher order applications of these technologies such as combining new energy generators with advanced desalination processes for fresh water production;**
- **Determination of the long-term social, cultural, and geo-political implications of these technologies as they help eliminate poverty, empower economies at the local/village level, and alter the zero-sum game of limited resources such as fossil fuels;**
- **Determination of medical, telecommunications and space travel applications of these technologies;**
- **Determination of methods to avoid and if necessary, contain the weaponization of such energy and propulsion technologies;**
- **Determination of the implications of these technologies for social justice, peace and the creation of a long-term sustainable civilization on Earth, among others.**

To accomplish the development of these policies and strategies is a tall order, but one that has to be carried out if we as a society plan to get off the “fossil fuel” economy and move to a more global sustainable life style. These policies and strategies must be developed simultaneously with the development of the actual technologies to assure acceptance and integration of the new technologies into our global society.

1.4. History of Over-Unity Technologies

Over-unity is a term often used to describe technologies, which use a closed system that produces more energy than it consumes. Technologically, the hurdles to achieve what is called over-unity energy generation or “energy from the vacuum”, by accessing the teeming energy in the space around us, are not insurmountable and have been well documented in the open literature. For example, Bearden (2002, 2007) lists well over a dozen potential over-unity technologies.

Research shows that over the past 75 years a number of significant breakthroughs in energy generation and propulsion have occurred that have been systematically suppressed. Since the time of N. Stubblefield, N. Tesla, T. Townsend Brown and others in the early and mid-twentieth century, we have had the technological ability to replace fossil fuel, internal combustion and nuclear power generating systems with advanced non-polluting electromagnetic and electrogravitic systems. Many of these breakthroughs are documented in recent books by Bearden (2002), Valone (2007) and Manning and Garbon (2008).

What has been insurmountable are the barriers created through the collusion of vast financial, industrial, oil and rogue governmental interests. In short, the strategic barriers to the widespread adoption of these new electromagnetic energy-generating systems far exceed the technological ones.

1.5. Categories of Suppression

Our review of past and now-obscure technological breakthroughs shows that these inventions have been suppressed or seized by the following broad categories of actions

- 1.5.1. Acquisition of the technology by 'front' companies whose intent has been to 'shelve' the invention and prevent the device from coming to market.
- 1.5.2. Denial of patents and intellectual property protection by systematic action by the US and other patent offices.
- 1.5.3. Seizure or suppression of the technology by the illegal application of section 181 of the US Patent law or other illegal applications of national security provisions that result in the technology being classified or deemed "of significance to the national security". Note that these applications are illegal actions taken by rogue, unsupervised individuals and entities who are working in collusion with interests to suppress these technologies.
- 1.5.4. Abuses by other regulatory or licensing entities, including but not limited to rogue elements within the Department of Defense, CIA, NSA, Federal Trade Commission, Department of Energy and others.

- 1.5.5. Targeting the inventor or company with financial scams, illegal financial arrangements that lead to the demise of the company, and similar traps.
- 1.5.6. Systematic interception of funds and essential financial support needed to develop and commercialize such fundamental new energy sources.
- 1.5.7. A pattern of harassment, bomb-threats, theft and other shadowy actions that frighten, intimidate and demoralize those inventing, holding or developing such technologies; significant bodily harm and murder have also apparently occurred.
- 1.5.8. Inducements through significant financial buy-outs, offers of positions of power and prestige and other benefits to the owner of such technologies to secure their cooperation in suppressing such technologies.
- 1.5.9. Scientific establishment prejudice and rejection of the technology in light of unconventional electromagnetic effects thought to be not possible by current scientific conventional wisdom.
- 1.5.10. Corruption of scientific entities and leaders through clandestine liaisons with rogue classified or shadowy private projects that intend to suppress such devices.
- 1.5.11. Corruption of major media entities and key figures through clandestine liaisons with rogue classified or private shadowy projects that intend to suppress such devices.

Considering the multiplicity of suppression techniques described above, a lone inventor or conventional R and D company can in no way overcome such obstacles. Therefore, a strategic plan and capability commensurate with these barriers must be devised and executed in order for these new technologies to succeed.

2. Strategic Plan Components to Safeguard Development

This brief overview illustrates the depth and breadth of the minimum strategy needed to move these technologies forward - and overcome the significant hurdles that exist in the status quo. The various components listed below must be carried out at the same time as the research and development phase in order to help ensure its success.

- 2.1. Use of intellectual property protection and patenting programs in the US and overseas that avoid triggering section 181 of the Patent Law and similar actions by rogue national security interests.
- 2.2. Scientific community education activities to create a core of mainstream and respected scientific and institutional support for this new science.
- 2.3. Political and governmental education activities to create firm support among political leaders, friendly government agencies, regulatory entities and others; an associated PAC (Political Action Committee) will need to provide support to those

- political leaders and groups that can in turn create real political currency in favor of these new energy systems.
- 2.4. Offshore (non-US) redundancy of technologies, resources and structural systems to ensure longevity in the event of catastrophic actions against the company or technology in the US.
 - 2.5. Mass media and public support creation by professional disclosure of the technology as soon as it is secured; such mass communications disclosure of the technology entails enlisting a top five PR firm and mass media release of scientific and government laboratory findings in support of the claims of the technology. This plan also requires the ability to complete a massive media buy of ad space in the event that the media refuses to carry the story (we have sources inside the media and the rogue national security structures that have confirmed that the media has been and is controlled on these and related issues and that it is prudent to plan for some degree of media obstruction). Some 70% of Americans are concerned with the future of the environment and sustainability issues and this majority of the population can become a powerful force of support should attempts to suppress the technologies occur.
 - 2.6. Careful liaison to existing mass movement groups and institutions that would naturally support the release and widespread adoption of such new energy systems; these include environmental organizations and agencies, anti-poverty groups and agencies, conservation and energy groups, international development groups and agencies, etc.
 - 2.7. Outreach to oil producing countries and interests to negotiate performance-based inclusion of their interests in the dissemination of these new technologies, thus 'hedging' the negative long-term economic impact of these technologies on their core economic foundation. The concept here is to make potential adversaries interested allies - but with specific performance criteria that would preclude the obstruction or 'shelving' of the technology.
 - 2.8. Education and development of further support within specific national security, military and intelligence arenas already friendly to the release of these technologies; a substantial level of support has been identified and can be further cultured in support of the safe and orderly release of these energy systems.
 - 2.9. The inventors of these technologies will be specifically insulated from such threats and pressures, legal or otherwise, since the technology will be assigned to the company via a professional licensing agreement that protects the inventor; the inventor is further assured that the device will be massively disclosed and disseminated by a clause that allows the technology to revert back to the inventor should the development company not perform under specific criteria within a specified time.

3. Organizational Needs to Develop Breakthrough Energy Technologies

An historical overview of over-unity electric generators and other unconventional energy innovations shows that for at least 75 years, human society could have had a replacement for fossil fuels and the internal combustion engine. The fact that such energy breakthroughs are not in use today is ample evidence that the hurdles to the widespread use of such technologies is less about inventing such a system - or even about adequate funding - and primarily about 'something else'.

Developing such a device - assuming it is robust, commercially viable and putting out real usable power - is less than 10% of the solution to this problem. The other 80-90% lies in this 'something else'. This deals with the special interests, cartels, power centers and the like that do not want a replacement for oil, coal, centralized public utilities and the internal combustion engine. These groups have enormous power to block the release of such a new energy technology and have used many of the suppression activities listed above.

A company and its core leadership must have a track record of taking on very controversial, suppressed areas of knowledge, and credibly moving them into the public arena of support and disclosure. For example, The Disclosure Project has a deep and wide area of expertise in dealing with so-called black and suppressed technology and information (see www.SiriusDisclosure.com). The Disclosure Project has a vast network of consultants, advisors, supporters and contacts from a wide range of disciplines, including the most senior levels of the scientific, financial, governmental, military, international, media and diplomatic communities.

The average - even the exceptional - inventor simply cannot marshal the enormous power, public support, political support, clout and protective services needed to successfully move an over-unity system from prototype to the world market. The Orion Project can do this.

This combination - bold, courageous leadership along with a vast network of powerful financial, public and government support - makes The Orion Project unique in the world today and the only really effective means of bringing these new energy technologies to the world.

This is that other 80% - that 'something else' - that has been missing for the past 75 years.

3.1. Important Restrictions for Anyone Developing These Technologies

3.1.1. They must not allow these technologies to be used for weapons development or weaponization and no technology should be licensed for any weapon applications.

3.1.2. The involvement of any funding source must be limited to 49% or less controlling interest in the technology for security reasons. In the past, financial and business interests have predictably 'sold out' to special interests in order to suppress such important and revolutionary energy developments. The developing entity must be firmly committed to fully

disclosing and making widely available these earth-saving technologies. No amount of funding should result in the reversal of this policy.

4. Example List of Technologies

The following list of candidate technologies is presented here includes *examples* of technologies that could be developed over a several-year period with the proper funding (note that several may take longer). Not all these technologies would need to be supported initially as there will have to be a very rigorous vetting of each technology during the initial stages to determine which ones have the best possibility of success. Estimated costs for simultaneous work on 6 of the projects are listed in the Example Budget section below (Section 5.0).

4.1. Solid-State Electromagnetic Free Energy Generator

This technology is truly revolutionary and extracts electromagnetic energy from the quantum vacuum field. The device is solid state, has no moving parts. The engineering team, with adequate funding support, will be able to create a multi-kilowatt freestanding free energy device within 12-18 months, ready for commercial application. The initial project objective will be the creation of portable power units, electronic power supply and internal units to use in power equipments and cabinets. These initial applications will be worldwide in scope. Later, home and business power units will be available. An example of this technology is described by Bearden (2002, Chapter 7).

4.2. Rotary Electromagnetic Free Energy Generators

These devices use a rotating electromagnetic field to extract energy from the quantum vacuum field. We have identified a number of inventors and companies working on their own variations of this technology. It is estimated that within one to two years with proper funding, one or several of these systems could be scaled to multiple kilowatts of usable output, adequate to run homes and small businesses.

4.3. Oscillated Power or Energy Resonance

Several inventors have developed Tesla's concept of oscillated power so that a standard alternating current (AC), which has alternating current and voltage, has, in addition, alternating power. This reduces the amount of power the utilities must "push" down the lines to do work, allowing the present utilities to increase available power to customers without the huge cost of new power plants. This technology would permit large electric power users to reduce consumption by up to 50%, and could be adapted for integration into consumer products to make them extremely efficient.

4.4. Electromagnetic Lifter Technology

This technology has been developed by a brilliant mathematician and physicist who has elucidated the fundamental relationship between electromagnetism, magnetic fields and mass and gravity. Essentially, this technology, already developed to the level of a proof of principle system, allows for a reduction in mass, allowing for lift. The principles involved allow for creating free energy generation systems as well as advanced aerospace applications. Examples of this technology have been described by Valone (1994, 2004), Cameron (2001) and LaViolette (2008).

4.5. Water Enhanced Combustion Technology

There are numerous technologies using electrolysis to dissociate water into a mixture of hydrogen and oxygen (often termed Brown's gas) that when added to the fuel stream in internal combustion engines reduces pollutants and "increases mileage". The main problem with most of these technologies is that the amount of energy necessary to dissociate the water is approximately equal to the amount of energy obtained during the combustion process, so there is no "energy gain". However, several inventors have investigated the physics of water dissociation and are able to break down water to Brown's gas using much less energy than physics predicts by using natural resonance frequencies of water dissociation. Both Dr. Andrija Puharich (1993) and Stanley Meyer (multiple patents, examples: 1986, 1989, 1992) developed ways to break water apart using these technologies.

4.6. Electrostatic Combustion Technology

This technology allows for a simple retrofit on existing internal combustion engines that reduces emissions by approximately 40% and increases fuel mileage by up to 20%. This inexpensive technology is applicable to diesel engines and large coal-fired industrial and electrical utilities, dramatically reducing acid rain and other pollution from existing power plants. (Note: more than 50% of US electricity is still produced by burning coal.) EPA labs and the Department of Transportation have evaluated this technology, and its efficacy has been proven.

4.7. Electromagnetic Free Energy Motor

We have acquired an expired patent, with complete drawings and specifications, for a free energy electromagnetic motor. This device was acquired over 20 years ago by financial and industrial interests that 'black – shelved' the technology.

4.8. Plasma Arc Reactor

We have identified an emerging company with proven plasma arc technology that completely reduces any organic, liquid waste material to a usable, clean-burning gas fuel. These reactors are already built and operating. This technology, if widely promoted and applied, would clean up and neutralize toxic waste sites, ‘Super-fund’ sites, treat radioactive waste material and other hazardous materials. This dual function – environmental cleanup and clean fuel generation- makes this technology an important part of the portfolio of emerging new energy systems.

4.9. Stubblefield Earth Battery and Communication Technologies

From the 1880s to the early 20th century, Nathan Stubblefield experimented with an electromagnetic induction wireless telephone that in his later research used natural conduction through the earth. He later (May 30, 1902), demonstrated this in Belmont Park, Philadelphia that has been memorialized in photograph of the attendees, which included Nicola Tesla. As Stubblefield continued his research he was able to power his system from the ground needing no external sources of power. The ground both provided the power and acted as the conductor of his signals. Unlike other researchers of the day, his system did not use spark gaps and it is suggested that the medium of the signal was not the normal electromagnetic waves (Hertzian waves) of today’s radio, but used either capacitance and later “earth magnetic waves”. Reports on his continued research into earth-powered technologies indicates that he was able to obtain both light and heat using the earth as a power source in addition to his communication technologies. This summary has been extracted from John Bedini’s website (Bedini, 2009) based in part on Vassilatos (1997). Note that recently concepts using non-electromagnetic fields for communication have been patented by Puthoff (1998).

4.10. Precursor Engineering Physics

Recently Thomas Bearden described some of the physics mechanisms that will allow the extraction of energy from the vacuum. He has written an easy-to-read briefing (Bearden 2009) entitled “Precursor Engineering and the Falsification of Modern Physics” which describes some history of energy physics and why energy science is at a stalemate today. A brief summary of this article is in Appendix B at the end of this document. It makes for very interesting reading and provides a clue as to why our energy sector is in such a mess and how we may get out of the dilemma. Included are a few of the far-reaching ideas and scientific references Bearden presents to get you intrigued. See the original article (Bearden 2009) to understand more fully the implications of what he is telling us.

He suggests that nothing less than “directly engineering physical reality at will” is possible using the Nobel laureate physicist Paul Dirac’s concept of “local vacuum tickling” (Dirac, 1928). This is obviously a potentially powerful use of physics that can be used to help in the extraction of zero point energy, but will require significant research to fulfill its promise.

4.11. Kowsky-Frost Antigravity Technology

In 1927 an article was published in the German journal, Radio Umschau (Anonymous, 1927a) describing research experiments by Prof. Kowsky and his engineer Mr. Frost. They discovered that if they placed a quartz crystal inside a metal box and treated it with high-powered radio waves the crystal expanded and caused the whole apparatus weighing 55 pounds to rise in the air. Later that year these results were published in a US magazine (Anonymous, 1927b) that is included in Appendix C. Research into these effects may lead to commercial anti-gravity or counter-gravity technologies that would revolutionize transportation throughout the world and need to be developed.

4.12. Other potential energy producing technologies

In addition to the above listed technologies, we at The Orion Project are working with over a dozen other technologies (some listed below) that with proper funding could be developed to the robust demonstration stage. *Furthermore, we expect that once The Orion Project or any other company has announced that they will be supporting Research and Development of over-unity type of energy technologies, there will be a flood of inventors coming forth with other technologies not listed.* We also expect that when these inventors realize that they have a safe place to continue their work, we may rapidly have access to already-developed “robust demonstrator” devices speeding up their commercial development. All technologies will need to be reviewed, to find those that appear to offer the best opportunity for success.

Some examples of these other possible technologies include:

1. A simple oscillating electronic circuit that puts out more energy as resistance heat than required to drive the device (nominally 1 watt in and greater than 10 watts out).
2. A solid state-switching device to pulse batteries with high voltage spikes to charge them to levels not normally obtainable by ordinary commercial charging technology (now commercially available).
3. A diode array solid-state chip, which refrigerates as it produces electrical power.
4. Several solid-state devices that extract energy from the quantum vacuum using either copper sheet/tube collectors or ceramic/metal materials on which are imposed very high frequency oscillations.
5. There are at least 6 different technologies that can be used to increase mileage (reduce fuel consumption) for gas burning generators and vehicles in addition to the ones described in the main section above.
6. Several different inventors have offered us over-unity rotary motor energy producing systems that need verification, further testing and development.
7. Commercialization of the scientifically proven phenomenon, the Bohren effect. This effect involves the use of low powered laser light input to a solution to which an optically scattering medium has been added.

8. A low powered solid-state battery using quantum crystal energy cells treated with high frequency plasma fields. Example prototypes presently exist.
9. Solid-state quantum cooling devices that require no energy input for long periods of time once they are charged, that can be applied to home or commercial refrigeration needs.
10. Commercialization of increased efficiency solar cells coupled with decreased fabrication costs to bring solar cell technology more in line with conventional technologies.

5. Example Budget Summary

This example budget summary is based on the assumption that there are 6 projects ongoing at the same time and not all projects will take 3 years to develop, so there will be overlap. It is for example purposes only.

SUMMARY 3 YEAR BUDGET	Year 1	Year 2	Year 3	3 Yr Costs
Research & Development Equipment/Materials				
General Lab Equipment	300,000	30,000	30,000	360,000
Project Specific Equipment	400,000	300,000	200,000	900,000
Machine Shop Equipment	175,000	25,000	25,000	225,000
Project Specific Materials	600,000	600,000	600,000	1,800,000
Total Equipment/Supplies Budget	1,475,000	955,000	855,000	\$3,285,000
Facilities Budget				
Lease (3 yr.) 9730 sq. ft. facility @ \$.80/sq. ft.	93,408	89,568	89,568	272,544
Leasehold Improvements -1 Location	250,000	30,000	20,000	300,000
Office Equipment (plus computers and software)	200,000	30,000	30,000	260,000
Utilities (5% inc/yr)	24,000	25,200	26,460	75,660
Telephone (5% inc/yr)	21,600	22,680	23,814	68,094
Internet Access (5% inc/yr)	7,200	7,560	7,938	22,698
Security Consulting, Systems, & Processes	400,000	400,000	400,000	1,200,000
Total Facilities Budget	996,208	605,008	597,780	\$2,198,996
Technical Development Team				
Mechanical Engineers (10) @85K/yr (5% inc/yr)	850,000	892,500	937,125	2,679,625
Electrical Engineers (10) @85K/yr (5% inc/yr)	850,000	892,500	937,125	2,679,625
Technical Assistants (8) @60K/yr (5% inc/yr)	480,000	504,000	529,200	1,513,200
Inventors (6) @150K/yr (5% inc/yr)	900,000	945,000	992,250	2,837,250
Machinist (1) @100K/yr (5% inc/yr)	95,000	99,750	104,738	299,488
Employee Taxes & Benefits (20%)	635,000	666,750	700,088	2,001,838
Total Technical Development Team Budget	3,810,000	4,000,500	4,200,525	\$12,011,025
Strategic and Administrative Expenses				
Administrative Salaries (5% inc/yr)	1,300,000	1,365,000	1,433,250	4,098,250
Admin. Support Salaries (5% inc/yr)	680,000	714,000	749,700	2,143,700
Consultants - Technical (12 @75000) (5% inc/yr)	900,000	945,000	992,250	2,837,250
Employee Taxes & Benefits (20%)	486,000	510,300	535,815	1,532,115
Science Advisory Board (honorarium) (5% inc/yr)	64,000	67,200	70,560	201,760
Licensing Fees (est. 6 inventors)	1,200,000	0	0	1,200,000

Consultants - Patent	100,000	150,000	150,000	400,000
Legal & Accounting	200,000	200,000	200,000	600,000
Subcontracted Technical Testing	100,000	200,000	200,000	500,000
Software Costs (upgrades, etc)	20,000	35,000	35,000	90,000
Supply Costs (5% inc/yr)	84,000	88,200	92,610	264,810
Publicity and Press Conference Costs	300,000	325,000	325,000	950,000
Media Advertising	500,000	500,000	500,000	1,500,000
Government Relations, Consulting/Planning	500,000	500,000	500,000	1,500,000
Travel Expenses (Staff) (5% inc/yr)	600,000	630,000	661,500	1,891,500
Travel Expenses (Advisory Boards) (5% inc/yr)	80,000	84,000	88,200	252,200
Misc. & Contingency Expenses (5% inc/yr)	1,000,000	1,050,000	1,102,500	3,152,500
Offshore Redundancy & Offices (5% inc/yr)	1,000,000	1,050,000	1,102,500	3,152,500
Total Administrative Costs	9,114,000	8,413,700	8,738,885	\$26,266,585
Total Example Budget for 3 Years				\$43,761,606

6. Appendix A. General References on Energy Technologies

This list is not meant to be thorough, but to guide the reader to books and articles of a broad general nature in the field of “zero point energy” or “free energy”. Some of the references include 100s of their own citations, while a search on Google results in millions of results.

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6.0 Appendix B. Precursor Engineering - Review of Issues

Recently Thomas Bearden described some of the physics mechanisms that will allow the extraction of energy from the vacuum. He has written an easy-to-read briefing (Bearden 2009) entitled “Precursor Engineering and the Falsification of Modern Physics” which describes some history of energy physics and why energy science is at a stalemate today. A brief summary of the physics is in the Appendix at the end of this document. It makes for very interesting reading and provides a clue as to why our energy sector is in such a mess and how we may get out of the dilemma. Below are a few of the far-reaching ideas and scientific references Bearden presents to get you intrigued. Please see the original article to understand more fully the implications of what he is telling us.

He suggests that nothing less than “directly engineering physical reality at will.” is possible using the Nobel laureate physicist Paul Dirac’s concept of “local vacuum tickling” (Dirac, 1928 Royal Society Series A, Vol. 117 (778), p. 610-624).

How might this happen? Well, in the classical concept of the “probabilities” vacuum, there are positive energy/positive probabilities of ever-higher energy interactions that ultimately result in the creation of every “observable” in the universe. Basically, the observable universe is “continually being created and ‘happening’ at an incredible rate” by these positive energy/positive probabilities.

Now, what if there was a way to produce some sort of negative energy/negative probabilities (negative E)?

Dirac describes the removal of the negative energy from physics: “One gets over the difficulty on the classical theory by arbitrarily excluding those solutions that have a negative E. One cannot do this in the quantum theory, since in general a perturbation will cause transitions from states with E positive to states with E negative.” [Dirac, Proc. of the Royal Society A, Vol. 117, p. 610-624]. Here he suggests that tiny perturbations will produce negative energy in the affected region of the vacuum. In other words, tiny-energy sharp-gradient pulses will add “negative probabilities” into a specific region so that highest “positive probabilities” that create and hold together a physical structure start to “unhappen”.

For example, if the correct sharp-gradient pulse were applied to water, the OH-bond positive-probability that holds water together would start to “unhappen” and the water molecules would just come apart as their OH bonds start to vanish statistically.

As our aware readers of The Orion Project may quickly realize, this suggests a powerful tool in the breaking apart of water to produce a combustible mixture of H and O or hydroxy (Brown's) gas. This is what might have been done by Stan Meyer, Dr. John Kanzius and other researchers, who used various pulsed electromagnetic energies to break apart water. They probably did this without an understanding of this potential mechanism. This mechanism of negative energy "unhappening", as Bearden describes it, can also be used in the "unhappening" of some diseases, another area of present research.

Finally, it is interesting to note that the concepts of "negative energy occur in the Schrödinger equation and in Dirac's relativistic electron theory, and so there it is in the very foundations of physics." Yet during the 1930's some physicists, afraid of the implications of this negative energy, removed as much of this knowledge as possible from classical physics. As noted by Hotson, "I think if one had to point to a single place where science went profoundly and permanently off the track, it would be 1934 and the emasculation of Dirac's equation." [D. L. Hotson, "Dirac's Equation and the Sea of Negative Energy," Part I, New Energy, Issue 43, 2002, pp. 1-20. Quote is from p. 1.]

Bearden calls the application of these processes "**precursor engineering**, since one is directly engineering the ongoing creative bubble-set of an observable entity, by not only "happening" the statistical processes involved in it but also by selectively "unhappening" the entire set or selected parts of it."

The ultimate application of such technology will allow "humankind to enter into the "final physics" and allow direct engineering of physical reality itself", hopefully for the overall betterment of mankind.

6.0 Appendix C. Description of the Kowsky-Frost Experiment

The following article was reprinted from the September, 1927 issue of *Science and Invention* (page 398) and copied from the following website with permission: <http://www.keelynet.com/gravity/KFrost.htm> . A version of this paper with the original figures can be found at: www.theorionproject.org/en/research.html .

Quartz Crystals Charged by High Frequency Current Lose their Weight

Although some remarkable achievements have been made with shortwave low power transmitters, radio experts and amateurs have recently decided that short-wave transmission had reached its ultimate and that no vital improvement would be made in this line. A short time ago, however, two young European experimenters working with ultra short-waves, have made a discovery that promises to be of primary importance to the scientific world.

The discovery was made about six weeks ago in a newly established central laboratory of the Nessartsadding-Werke in Darredein, Poland, by Dr. Kowsky and Engineer Frost. While experimenting with the constants of very short waves, carried on by means of quartz resonators, a piece of quartz which was used, suddenly showed a clearly altered appearance. It was easily seen that in the center of the crystal, especially when a constant temperature not exceeding ten degrees C. (50 degrees F.) was maintained, milky cloudiness appeared which gradually developed to complete opacity.

The experiments of Dr. Meissner, of the Telefunken Co., along similar lines, according to which quartz crystals, subjected to high frequency currents clearly showed air currents which led to the construction of a little motor based on this principle. A week of eager experimenting finally led Dr. Kowsky and Engineer Frost to the explanation of the phenomenon, and further experiments showed the unexpected possibilities for technical uses for the discovery.

Some statements must precede the explanation. It is known at least in part, that quartz and some other crystals of similar atomic nature, have the property when exposed to potential excitation in a definite direction, of stretching and contracting; and if one uses rapidly changing potentials, the crystals will change the electric waves into mechanical oscillations.

This piezoelectric effect, shown in Rochelle salt crystals by which they may be made in sound producing devices such as loudspeakers, or reversely into microphones, also shows the results in this direction. This effect was clearly explained in August, 1925 Radio News and December, 1919 Electrical Experimenter. These oscillations are extremely small, but have nevertheless their technical use in a quartz crystal wave-meter and in maintaining a constant wavelength in radio transmitters. By a special arrangement of the excitation of the crystal in various directions, it may be made to stretch or increase in length and will not return to its original size.

It seems as if a dispersal of electrons from a molecule resulted, which, as it is irreversible, changes the entire structure of the crystal, so that it cannot be restored to its former condition.

The stretching out, as we may term this strange property of the crystal, explains the impairment of its transparency. At the same time a change takes place in its specific gravity. Testing it on the balance showed that after connecting the crystal to the high tension current, the arm of the balance on which the crystal with the electrical connections rests, rose into the air.

This pointed the way for further investigation and the determination of how far the reduction of the specific gravity could be carried out. By the use of greater power, finally to the extent of several kilowatts and longer exposure to the action, it was found eventually that from a little crystal, 5 by 2 by 1.5 millimeters, a non-transparent white body measuring about ten centimeters on the side resulted, or increased about 20 times in length on any side.

The transformed crystal was so light that it carried the whole apparatus with itself upwards, along with the weight of twenty-five kilograms (55 lbs.) suspended from it and floating free in

the air. On exact measurement and calculations, which on account of the excellent apparatus in the Darredein laboratory, could be readily carried out, it was found that the specific gravity was reduced to a greater amount than the change in volume would indicate. Its weight had become practically negative.

There can be no doubt that a beginning has been made toward overcoming gravitation. It is to be noted, however, that the law of conservation of energy is absolutely unchanged. The energy employed in treating the crystal, appears as the counter effect of gravitation. Thus the riddle of gravitation is not fully solved as yet, and the progress of experiments will be followed further.

It is, however, the first time that experimentation with gravitation, which hitherto has been beyond the pale of all such research, has become possible, and it seems as if there were a way discovered at last to explain the inter-relations of gravity with electric and magnetic forces, which connection, long sought for, has never been demonstrated. This report appears in a reliable German journal*, "Radio Umschau."

*NOTE: The reference for the Radio Umschau article referred to above is: Ueberwindung der Schwerkraft? Ein neuer Erfolg der Quarzkristallforschung. Radio Umschau. pp. 218-220. April 1, 1927.